

AR 524: Robot Simulators

Course Code: AR 524

Course Name: Robot Simulators

L-T-P-C: 2-0-2-3

Prerequisites: Linear Algebra (IC 114) or equivalent courses, and basic programming skills

Intended for: B.Tech, M.Tech, M.Tech. (R), PhD

Mutual Exclusion: None

Approval: 48th Senate (listed in the B.Tech General Engineering curriculum); syllabus approved in 64th BoA

Preamble

This course provides a hands-on, concept-driven learning to robot simulation ecosystems, bridging theoretical modeling with industry-relevant simulation workflows. Students will learn why simulators are indispensable, how robot motion and sensors are mathematically modeled, and simulated inside state-of-the-art robotic simulators.

By integrating robot motion and sensor models for state estimation along with ROS-based programming, the course equips students with the practical skill set required to transition seamlessly from simulation to real-world deployment. Exposure to modern simulation platforms and workflows—including synthetic data generation, hardware-in-the-loop testing, and digital twin creation—ensures strong alignment with current industry practices.

By the end of the course, students will acquire the knowledge and skills required to understand the role of high-fidelity digital twins, design realistic simulation environments, and validate algorithms before hardware deployment, making them well-prepared for careers in robotics, Physical AI, and advanced industrial automation.

Course Modules

- **Foundations and Motivation of Robot Simulation:** Why Robot Simulators? Concepts, advantages, types, Key Components of Robot Simulators and Selection Criteria, Role of simulation in robotics design and validation, Synthetic Data Generation for Physical AI and Scaleup the simulation environment using Gen2Sim. (4 Hours Theory + 2 Hours Lab)
- **Robot Motion Modeling and Simulation:** Unicycle, Bicycle model, Robot Jacobian and Quadcopter model, Velocity motion model, Odometry motion model and their simulation. (6 Hours Theory + 6 Hours Lab)
- **Robot Programming With ROS:** Fundamentals, Architecture for ROS1 and ROS2, Publisher and Subscriber node, rostopic, Rviz, URDF Modeling and ROS Navigation Stack. (6 Hours Theory + 6 Hours Lab)
- **Sensor Modeling, Calibration, Filtering, and Data Fusion:** Beam range finder model, Camera model and sensor calibration, EKF and Quantum stochastic filtering. (6 Hours Theory + 6 Hours Lab)
- **Popular Robot Simulation Platforms and Digital Twins:** ROS–Gazebo integration, PyBullet for manipulation and control, AirSim for aerial robotics, CARLA for autonomous vehicles, Isaac Sim for AI-driven robotics, Environment modeling, Sim2Real transfer considerations, HIL, and digital twins. (6 Hours Theory + 8 Hours Lab)

Text Books

1. Thrun, Sebastian, *Probabilistic robotics*, Communications of the ACM 45.3 (2002): 52-57.
2. Corke, Peter I., Witold Jachimeczyk, and Remo Pillat, *Robotics, vision and control: fundamental algorithms in MATLAB*, Vol. 73. Berlin: Springer, 2011.

References

1. Katara, Pushkal, Zhou Xian, and Katerina Fragkiadaki, *Gen2sim: Scaling up robot learning in simulation with generative models*, IEEE International Conference on Robotics and Automation (ICRA), 2024.
2. Quigley, Morgan, Brian Gerkey, and William D. Smart, *Programming Robots with ROS: a practical introduction to the Robot Operating System*, O'Reilly Media, Inc., 2015.
3. Joseph L., *Mastering ROS for Robotics Programming: Design, build, and simulate complex robots using the Robot Operating System*.
4. Collins, Jack, et al., *A review of physics simulators for robotic applications*, IEEE Access 9 (2021): 51416-51431.
5. Behera, Laxmidhar, and Indrani Kar, *Quantum stochastic filtering*, IEEE International Conference on Systems, Man and Cybernetics. Vol. 3, 2005.
6. Fairchild, Carol, and Thomas L. Harman, *ROS robotics by example*, Packt Publishing Ltd, 2016.
7. Coumans, Erwin, and Yunfei Bai, *PyBullet quickstart guide*, ed: PyBullet Quickstart Guide. <https://docs.google.com/document/u/1/d> (2021).